SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I Naveen Kumar S - 192421030 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naveen Kumar S

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Answers :

1. L1, L2, and L3 Cache Comparison

- **L1 Cache:** Smallest and fastest cache; has the **lowest latency** and very high bandwidth. It provides the highest throughput for frequently used data.
- **L2 Cache:** Larger but slower than L1. It has slightly higher latency and lower bandwidth, but still provides good throughput.
- **L3 Cache:** Largest and slowest cache, usually shared among CPU cores. It has the highest latency and lower bandwidth compared with L1/L2.
- **Impact:** A higher cache hit rate reduces access to slower main memory, improving overall **CPU performance and throughput**.

2. SRAM vs DRAM

- **SRAM:** Faster access, very low latency, and high bandwidth/throughput. It is expensive and requires more space, so it is mainly used for **CPU cache**.
- **DRAM:** Higher latency and generally lower access performance than SRAM, but it offers high storage density at lower cost. It is mainly used as **main memory (RAM)**.
- **Conclusion:** SRAM improves processor speed, while DRAM provides large-capacity memory economically.

3. Virtual Memory vs Paging vs Cache Memory

- **Cache Memory:** Very low access latency and high throughput because it is close to the CPU.
- **Paging/Virtual Memory:** Uses secondary storage to extend available memory. It has much higher latency than cache and can reduce system throughput when page faults occur.
- **Comparison:** Cache improves CPU performance by reducing memory access time, while virtual memory improves memory capacity but may decrease performance due to slow disk/SSD accesses.

4. NAND Flash vs DRAM

- **DRAM:** Very low latency, high bandwidth, and high throughput. It is suitable for **main memory** where frequent read/write operations occur.
- **NAND Flash:** Much higher latency than DRAM, but it provides high storage density, non-volatility, and good sequential bandwidth.
- **Conclusion:** DRAM is used for fast working memory, while NAND Flash is used for **persistent storage** such as SSDs.

5. MRAM vs RRAM

- **MRAM:** Offers fast access, good throughput, non-volatility, and relatively high endurance. It can provide good energy efficiency because it does not require continuous refresh.
- **RRAM:** Uses resistance states to store data and can offer low-power operation, high density, and good scalability. Its access time and endurance depend on the specific implementation.
- **Conclusion:** MRAM is attractive for **fast, reliable memory**, while RRAM is promising for **high-density and energy-efficient memory systems**.